

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Semester Examination - Summer 2023 Course: B. Tech. Branch: Computer Engineering Semester: VII Subject Code & Name: BTCOC601 – Compiler Design Max Marks: 60 Date: 13/ 07 / 2023 Duration: 3 Hrs.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1 Solve Any Two of the following.			
A)	Explain the various phases of a compiler and their roles in the compilation process. Illustrate the flow of information between these phases.	Remember	6
B)	Discuss the lexical analysis phase of a compiler. Explain the role of the lexical analyzer and the construction of tokens from the input source code.	Remember Understand	6
C)	What is syntax analysis? Describe the role of a parser in this phase. Differentiate between top-down and bottom-up parsing techniques, providing examples for each.	Application Remember	6
Q.2 Solve Any Two of the following.			
A)	Explain the concept of a symbol table in compiler design. Discuss its importance and the information it stores during the compilation process.	Remember Understand	6
B)	Describe the different types of errors that can occur during compilation. Explain the difference between lexical, syntax, and semantic errors. Provide examples for each.	Remember Understand	6
C)	Explain constructing syntax trees for simple expressions involving only binary operators + and -. State the use of <i>Leaf</i> and <i>Node</i> in this syntax tree.	Understand Apply	
Q. 3 Solve Any Two of the following.			
A)	Discuss the role of semantic analysis in a compiler. Explain how type checking and type inference are performed during this phase. Illustrate with suitable examples.	Remember Understand	6
B)	Explain the intermediate code generation phase of a compiler. Describe the different intermediate representations used and their advantages and disadvantages.	Knowledge Understand	6
C)	What is optimization in the context of a compiler? Discuss the importance of optimization and provide examples of common optimization techniques used in compilers.	Knowledge Understand	6
Q.4 Solve Any Two of the following.			
A)	Describe the code generation phase of a compiler. Explain the different code generation techniques, such as the use of registers and stack allocation.	Understand Application	6
B)	What are the different types of code optimization techniques? Explain loop optimization and its benefits in detail. Provide examples to illustrate your explanation	Application Understand	6
C)	Construct a Predictive parsing table for the Grammar $E \rightarrow E+T \mid T, T \rightarrow T * F \mid F, F \rightarrow (E) \mid id.$	Understand Apply	6

.Q. 5	Solve Any Two of the following.		
A)	Explain the concept of runtime environments in a compiler. Discuss the stack allocation and heap allocation strategies used to manage memory during program execution.	Analyze Understand	6
B)	Discuss the concept of code generation for control flow statements in a compiler.	Remember Understand	6
C)	Explain how the assignment statement " <i>position = initial + rate * 60</i> " is grouped into the lexemes and mapped into the tokens passed on the syntax analyzer.	Understand Apply	6
	*** End ***		